Sulav Sapkota

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Education

Washington and Lee University Expected May 2029

Intended Major: B.S. in Mathematics and Computer Science — **GPA:** 3.93/4.00 *Lexington, VA*

- **Honors:** President's List 2025–2026
- **Pre-College Honors:** Gold Medal, National Chemistry Olympiad 2023; National Dynamite, National Olympiad of Informatics 2022; Gold Medal, Hydrocon 2.0 2022 (built a maze-solving robot using Arduino)
- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Intro to CS, Data Structures and Algorithms

Technical Skills

Languages & Tools: Python, C++20, R, SQL, JavaScript, React, Node.js, Django, Git/GitHub, Make, Unit Testing
ML and Geospatial: TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, geopandas, rasterio, QGIS
Finance: DCF, Comparable Company Analysis, Financial Modeling

Experience

Washington and Lee University June 2026 – Present

Machine Learning Research Assistant *Lexington, VA*

- Built an end-to-end Python pipeline to predict the protected side of aerial fire-retardant drops from burn maps
- Processed 21,000+ drop polygons from 424 US wildfires with geopandas and rasterio in a reproducible workflow
- Engineered raster-split, unburned-distance and expanded-polygon features, boosting accuracy from 81% to 91%
- Evaluated Random Forest and XGBoost with fire-grouped 5-fold CV to avoid leakage, reaching 91.5% accuracy

Kathmandu Institute of Applied Science June 2023 – June 2025

Machine Learning Researcher *Lalitpur, Nepal*

- Built U-Net and VGG16 models in TensorFlow to segment and classify microplastics in stereo-microscope images
- Reached 82% test accuracy on binary detection (783 images) and extended the models to classify 5 plastic types
- Added YOLOv8 detection (0.40 mAP50) and OpenCV post-processing to count particles against ground truth
- Won Best Poster at MoChWo 2024, an international conference in Kathmandu, presenting to 100+ researchers

Water Stories of Patan June 2024 – August 2024

Student Researcher *Lalitpur, Nepal*

- Managed a 6-member team's digital water-heritage project in Patan, mentored by a City Planning student at MIT
- Surveyed 30+ locals across 10+ neighborhoods; interviewed NGOs, INGOs and agencies using snowball sampling
- Mapped water policy (1963–2015) and stakeholders, and ran a conservation workshop with Patan middle-schoolers
- Curated findings on Patan's traditional water sources, student sentiment and contamination for a public website

Hello World I.T Solutions January 2021 – June 2024

Software Developer *Kathmandu, Nepal*

- Built a React frontend and Django REST API serving fixtures for an Australia-based, Nepali-owned soccer team
- Built a React/Node.js QI Dashboard with Chart.js visualizing 12 health KPIs across 7 provinces; live in production
- Added an interactive map of Nepal to the dashboard that users can click and zoom to explore individual districts
- Built the React.js frontend of an AI-featured mental health patient-tracking portal with a focus on data privacy

Projects

Himalayan Quake Atlas | *Python, SQLite, pandas, Chart.js, Leaflet*

- Built a Python and SQLite pipeline that fetches, cleans and stores 10,125 USGS earthquakes from 1990 to 2026
- Wrote 16 pytest unit tests covering data cleaning, the database, and b-value recovery on synthetic catalogs
- Estimated the Gutenberg-Richter b-value (1.25 ± 0.03) and showed it swings from 0.79 to 1.27 with the cutoff
- Shipped a Chart.js/Leaflet dashboard on GitHub Pages; regional b-values proved statistically indistinguishable

Rumor Mill | *C++20, Python, pandas, matplotlib*

- Built a C++20 discrete-event gossip simulator with a priority-queue event engine, seeded RNG, loss and latency
- Wrote 6 unit tests (graph connectivity, time limit, same-seed reproducibility, loss extremes) and a Makefile build
- Ran 180 simulations on a 1000-node graph; raising fanout from 1 to 6 cut time to 99% coverage from 10.6 to 1.8
- Found total messages sent stayed ∼10–11k across fanouts, with 85–90% duplicates: fanout buys speed, not lower cost

Leadership / Extracurricular

Williams Investment Society January 2026 – Present

Shadow *Lexington, VA*

- Selected as 1 of 10 first-year members of the WIS shadow program, a student club managing a $30M+ portfolio
- Pitched GOOGL and VISA holdings using DCF, comparable company analysis, and qualitative sector research